Contents

1. Author and Contact Information 2
2. Prerequisites 2
3. Installation of Application Files 2
4. Running the application 2
5. Access data from OPC UA Client 3
6. Validate if the Client Supports Required OPC UA Features 3
7. Tightening System Entry Point - AddressSpace View 3
8. Asset Data 4
9. Result Data 6
10. Result, RequestedResult Variable and Result Simulation Methods 7
11. Result Data Access View – Subscribed Result Variable Data Example 8
12. Subscribe to Result Events or Other Events 9
13. Event Data 10
14. Commands Example 11

1. Author and Contact Information

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    - **Editor** of VDMA OPC UA Industrial Joining Technologies Working Group.
- Contact for any questions/updates/support on using the demo and extending it.

2. Prerequisites

- **Windows Binary**
    - Windows 10 or later (Built using Windows SDK Version: **10.0.26100**).
    - Download from the following link: Windows_SDK_Dowload
    - Download **Visual Studio 2022 Redistributable**: VC-Redist_Download
- **Docker Image: Ensure** that Docker is installed and running.
- **OPC UA Test Client:** Download and install any OPC UA Client. **Example**: UaExpert_Download
- **IJT CS Reference:**
    - OPC_40450_Joining_Base
    - OPC_40451_Tightening

3. Installation of Application Files

- Download the following files in the **Installation Directory: OPC_UA_IJT_Server_Simulator**.
    - **opcua_ijt_demo_application.exe**
    - Contains **multiple** NodeSet files in XML format as below:
        - Opc.Ua.XXX.NodeSet2.xml
    - **Optional Files**
        - **Multiple JSON files** for **simulation**
            - server_configuration.json
            - simulated_asset_data.json
            - …
        - Dockerfile

4. Running the application

- **Common Steps**
    - **Go** to the **"OPC_UA_IJT_Server_Simulator"** directory.
    - The **EndpointUrl** of the OPC UA Server would be:
        - **opc.tcp://localhost:40451** or **opc.tcp://YourComputerName:40451**.
- **Windows Binary**
    - Ensure that the user has **Read/Write access** to the **Installation Directory**.
    - **Launch** the binary file (**opcua_ijt_demo_application.exe**).
        - Run as Administrator or at least with **Read/Write** access.
- **Docker Image**
    - **Run** the following commands which will run the simulator in a docker container:
        - docker build -t opcua_ijt_demo_application .
        - docker run --rm -p 40451:40451 opcua_ijt_demo_application

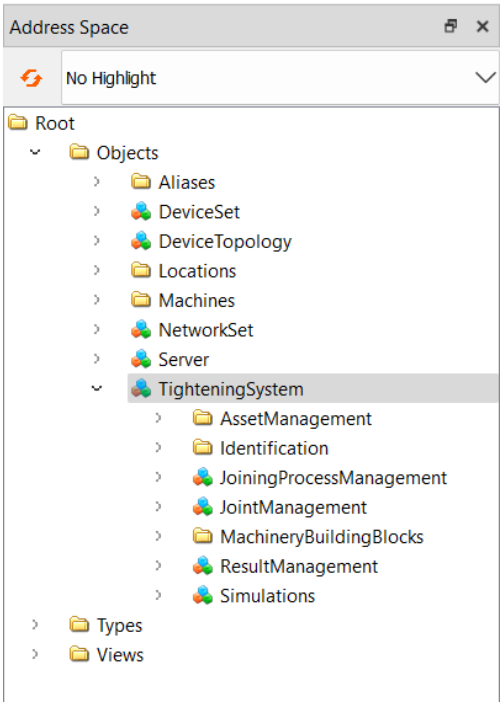
5. Access data from OPC UA Client

- Launch the OPC UA Client and connect to the given **EndpointUrl**.
- It will show the primary entry point: **TighteningSystem.**
- **All** the **Nodes** shown below are as per the **Companion Specification.**
- The **Simulations** node is the Application Node.

6. Validate if the Client Supports Required OPC UA Features

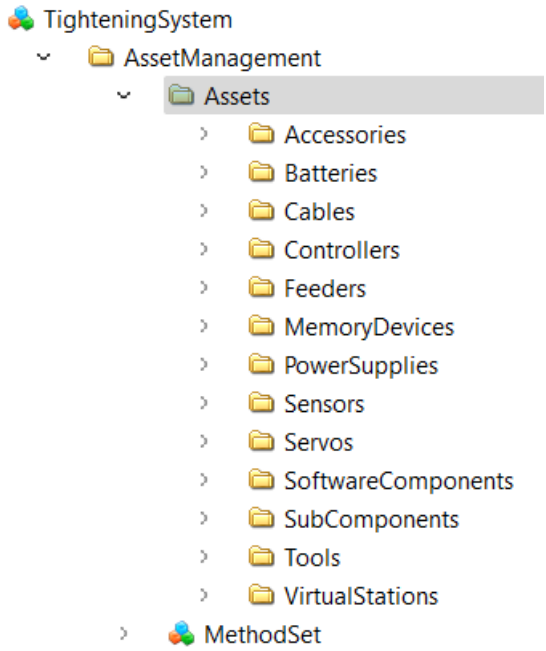
- OPC UA Client **shall support** OPC UA **Extension** Objects (Custom Structure Types) to **consume** the data as per the OPC UA IJT Standard.
- The quick test to validate if the Client application supports **extension** objects is to subscribe to **Result variable** or **Result Event** and visualize if the data is readable from the Client.
- Refer to **upcoming** sections on how to **subscribe** to **Result** variable or **Result** Event.

7. Tightening System Entry Point - AddressSpace View

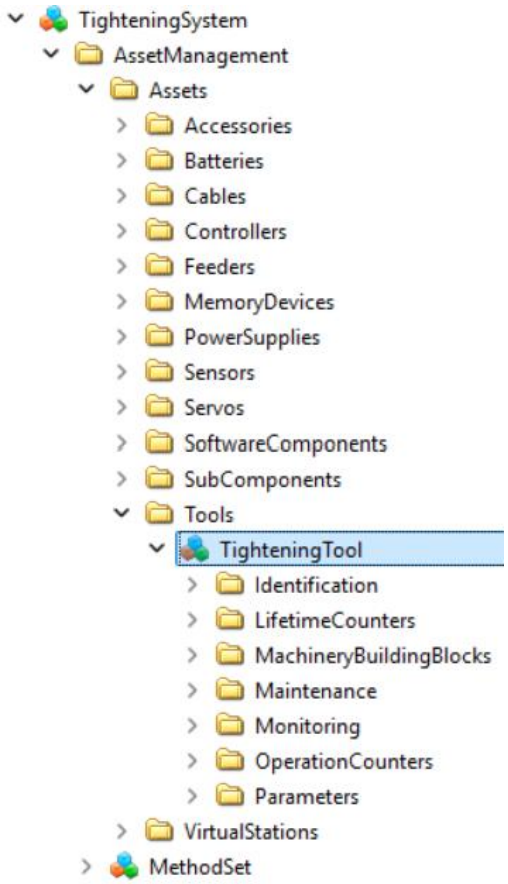


8. Asset Data

Browse the respective Asset Nodes from the address space and subscribe/read the respective data. **Drag** and **Drop** the **variables** to subscribe in the **Data Access View** window.



Example Asset Address Space View

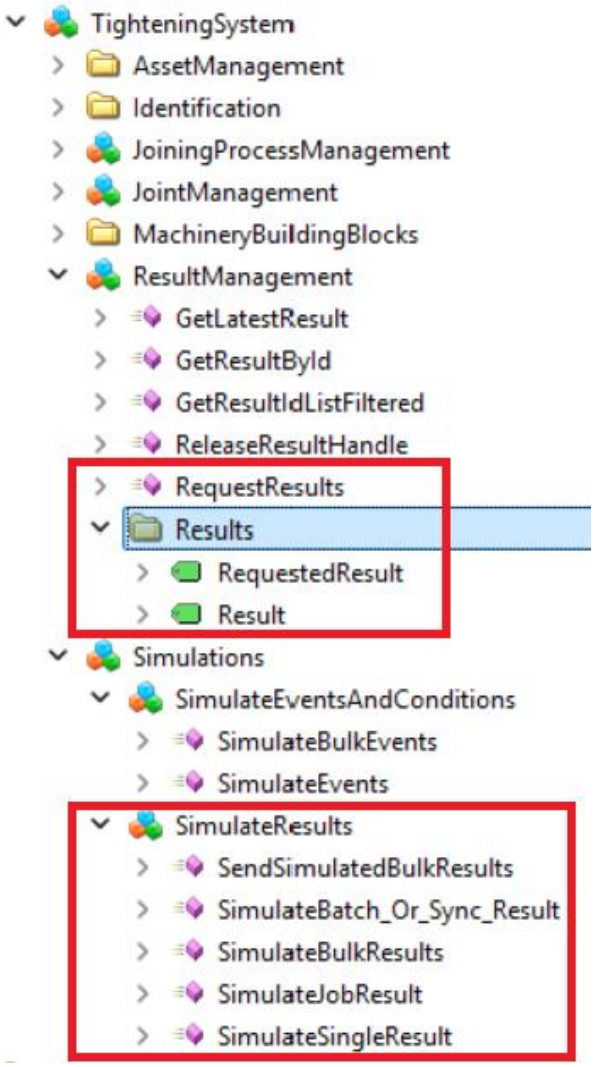

TighteningSystem
AssetManagement
Assets
Accessories
Batteries
Cables
Controllers
Feeders
MemoryDevices
PowerSupplies
Sensors
Servos
SoftwareComponents
SubComponents
Tools
TighteningTool
Identification
LifetimeCounters
MachineryBuildingBlocks
Maintenance
Monitoring
OperationCounters
Parameters
VirtualStations
MethodSet

9. Result Data

- **Result Access Options:**
  - Subscribe to the **Result** or **RequestedResult variable** shown below. **Drag** and **Drop** the **variables** to subscribe in the **Data Access View** window.
    - **Result** Variable is intended for getting **live** results.
    - **RequestedResult** Variable is intended for getting **historical** results.
  - Subscribe to **events** by subscribing to the **Server** node in the Event View.
- **Simulation Options**
  - Use the following **three methods** to simulate different types of **Results**. A new **Result** is generated upon the execution of the following **methods**.
    - **SendSimulatedBulkResults**
      - **To send already generated bulk results without recreating.**
    - **SimulateBatch_or_Sync_Result**
      - To generate Batch Result.
    - **SimulateBulkResults**
      - To generate several live Results.
    - **SimulateJobResult**
      - To generate a Job Result.
    - **SimulateSingleResult**
      - To generate a single live Result.
  - The **simulated** data is similar to the examples **defined** in the **Annexure** sections of the Companion Specification.

10. Result, RequestedResult Variable and Result Simulation Methods

Subscribe to Result variable as shown below.



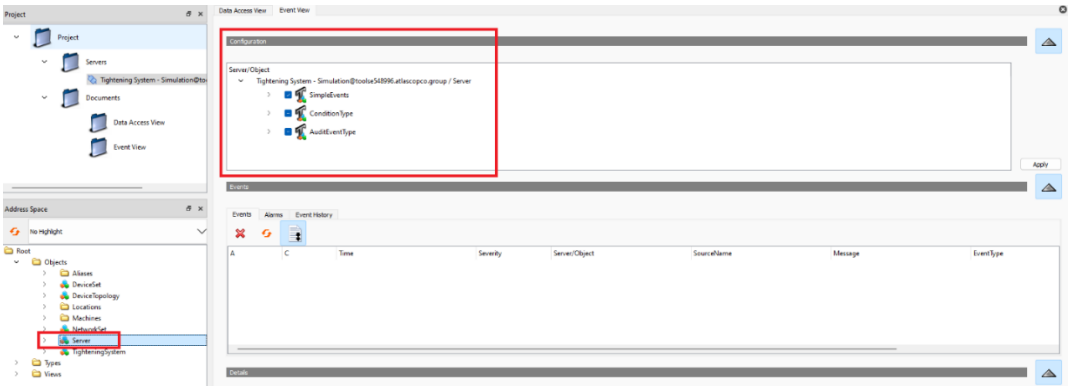
11. Result Data Access View – Subscribed Result Variable Data Example

The outcome of the **Result variable subscription** is shown below.

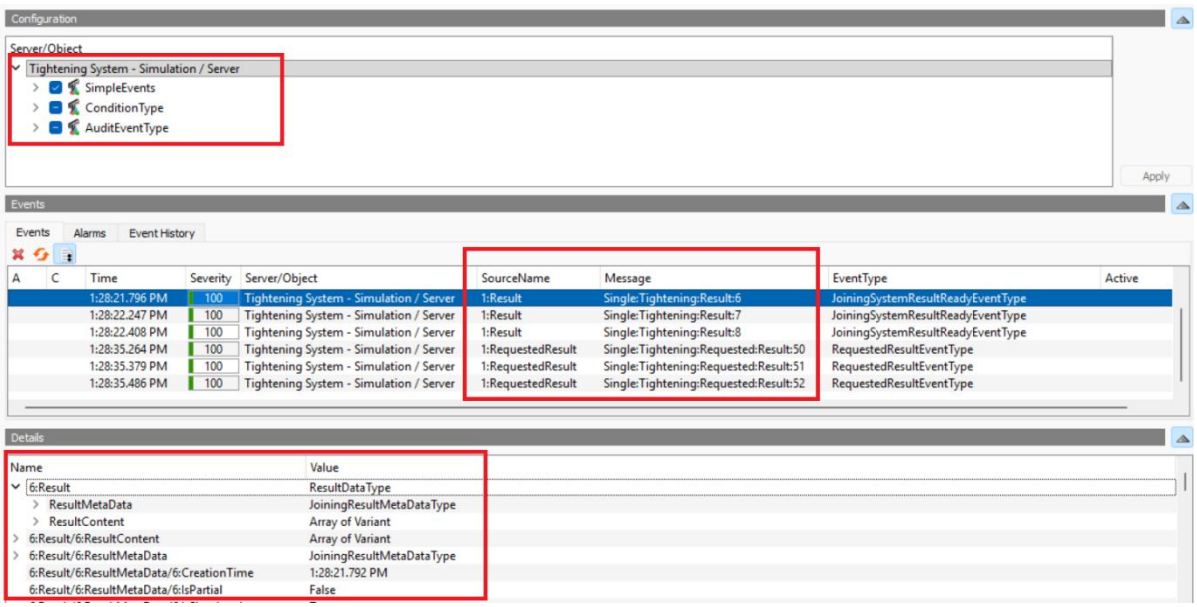
#	Display Name	Value	Datatype	Source Timestamp
1	Result	Double click to display value	ExtensionObject	8:35:14.750 PM
2	ResultContent	Double click to display value	Variant	8:35:14.752 PM
3	ResultMetaData	Double click to display value	ExtensionObject	8:35:14.753 PM
4	AssemblyType	1	Byte	8:35:14.753 PM
5	AssociatedEntities	Double click to display value	ExtensionObject	8:35:14.753 PM
6	Classification	1	Byte	8:35:14.753 PM
7	CreationTime	2025-10-09T18:35:04.654Z	DateTime	8:35:14.753 PM
8	Description	"en", "Single:Tightening:Result:2"	LocalizedText	8:35:14.753 PM
9	ExtendedMetaData	Double click to display value	ExtensionObject	8:35:14.753 PM
10	InterventionType	0	Byte	8:35:14.753 PM
11	IsGeneratedOffline	false	Boolean	8:35:14.753 PM
12	IsPartial	false	Boolean	8:35:14.753 PM
13	IsSimulated	true	Boolean	8:35:14.753 PM
14	JoiningTechnology	"en", "Tightening"	LocalizedText	8:35:14.753 PM
15	Name	Single:Tightening:Result:2	String	8:35:14.753 PM
16	OperationMode	2	Byte	8:35:14.753 PM
17	ProcessingTimes	Double click to display value	ExtensionObject	8:35:14.753 PM
18	ResultCounters	Double click to display value	ExtensionObject	8:35:14.753 PM
19	ResultEvaluation	1 (OK)	Int32	8:35:14.753 PM
20	ResultEvaluationCode	0	Int64	8:35:14.753 PM
21	ResultEvaluationDetails	"en", "OK TIGHTENING"	LocalizedText	8:35:14.753 PM
22	ResultId	15EC5500-CBA9-F648-A323-30BA63754435	String	8:35:14.753 PM
23	ResultState	1	Int32	8:35:14.753 PM
24	SequenceNumber	2	UInt64	8:35:14.753 PM

12. Subscribe to Result Events or Other Events

- Connect to the OPC UA Server using UaExpert or any other OPC UA Client.
- Subscribe to "**Server**" Object. In UaExpert, Drag and Drop the "**Server**" Object in the following **Configuration Window** as shown below.



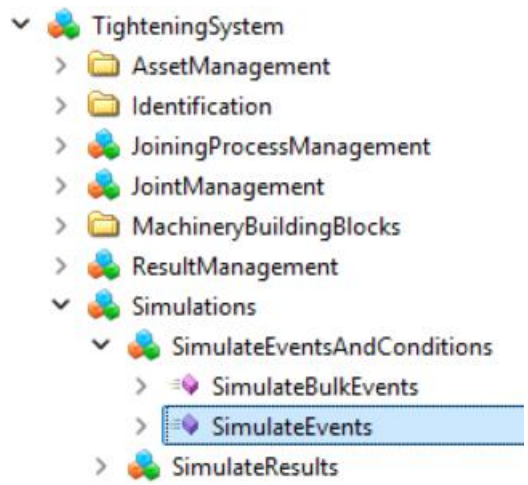
- Select the "**Simple Event**" **checkbox** and it shall **show** as checked: ✓ and Click on **Apply**.
- Generate a **new Result**, and the Result will be listed in the **Events Window** as shown below.



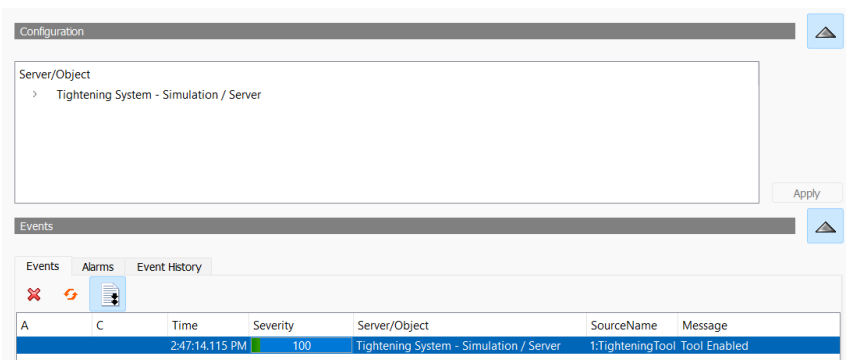
13. Event Data

Invoke the **SimulateEvents** or **SimulateBulkEvents method** as shown below to **generate 50+** different type of events.

Note: The **content** of the Events would be similar to any type of event from a joining system.



Example Events View

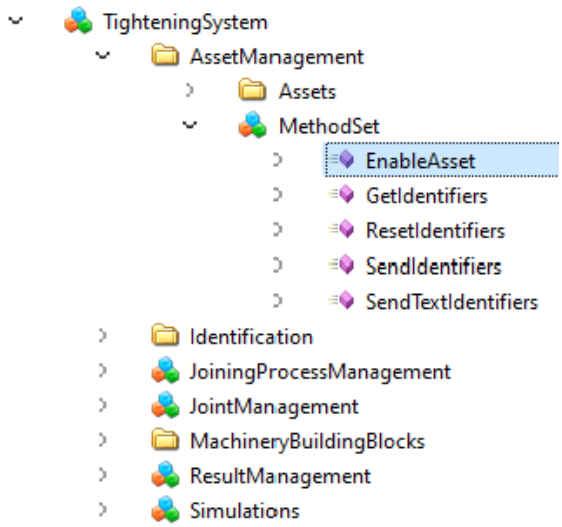
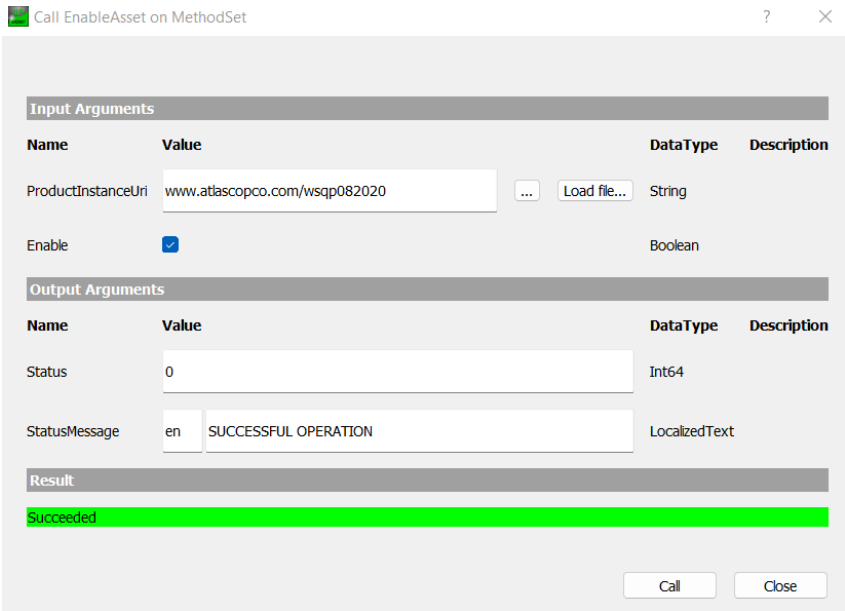


3:JoiningSystemEventContent/3:AssociatedEntities	Array of ExtensionObject
[0]	EntityDataType
Name	ProductInstanceUri
Description	ProductInstanceUri of the Asset
EntityId	www.atlascopco.com/wsqp082020
EntityOriginId	Null
IsExternal	True
EntityType	4
3:JoiningSystemEventContent/3:EventCode	0
3:JoiningSystemEventContent/3:EventText	"en", "Tool Enabled"
3:JoiningSystemEventContent/3:JoiningTechnology	"en", "Tightening"
3:JoiningSystemEventContent/3:ReportedValues	Array of ExtensionObject
ConditionClassId	NodeId
NamespaceIndex	0
IdentifierType	Numeric
Identifier	11166 [SystemConditionClassType]
ConditionClassName	"en", "SystemConditionClassType"
ConditionSubClassId	Array of NodeId
[0]	NodeId
NamespaceIndex	3
IdentifierType	Numeric
Identifier	1024
ConditionSubClassName	Array of LocalizedText
[0]	"en", "AssetEnabledConditionClassType"
EventId	len=1, 0x39

14. Commands Example

An example simulation of **EnableAsset** is provided. It takes the input of the ProductInstanceUri of the Tool.

Few error cases can be simulated when the input argument is empty or invalid. A respective error is shown in the output arguments.



Example Command View